

# Nestlé Insight

## Final System Clarification Document

*Updated after lecturer discussion - team reference for use cases, activity diagrams, testing, and development*

|                          |                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>           | Single shared reference so every group member follows the same final plan.                                                                                                                              |
| <b>Audience</b>          | Use case diagram team, activity diagram team, development team, and testing team.                                                                                                                       |
| <b>Project Direction</b> | The revised plan expands the original sales-rep-focused concept into a broader territory execution platform with six user roles.                                                                        |
| <b>Current Status</b>    | This document reflects the latest changes agreed in the chat, including Territory Distributor, shop assignment by Sales Representatives, warehouse-controlled allocation, and a lighter Shop Owner app. |

## 1. Why the project exists

Nestlé Insight exists to close the traditional-trade data gap. The original proposal explains that small grocery shops do not use POS systems, so their daily sales and stock movement are invisible to digital planning tools. The system is meant to act as a digital bridge between field operations and forecasting tools such as ARS and Python-based analytics.

Design principle: the system should improve real operational visibility without trying to replace Nestlé's existing forecasting platforms.

## 2. What changed from the original proposal

The original proposal mainly focused on Sales Representatives using a mobile app and planners/admin users using a web dashboard. After the lecturer discussion, the plan was expanded and reorganized into a more role-based execution model.

- The system now has six roles instead of a simpler two-layer rep/planner split.
- Warehouse Managers now control the start-of-day stock issue and product transaction records.
- Territory Distributors were introduced as a separate operational role for assigned deliveries.
- Sales Representatives no longer enter carrying stock manually at the start of the day; warehouse-issued stock drives the lorry dashboard.
- Shop Owners now have a lightweight ordering app instead of the earlier heavier feature set.
- Demand Planners are limited to ARS integration, non-POS insights, and forecasting-related features.
- The final plan now distinguishes who creates orders, who fulfills them, and who monitors them.

## 3. Final system vision

The final system should be understood as a territory execution and demand-signal platform. It supports field selling, territory delivery, warehouse stock movement, shop ordering, administrative control, and demand planning outputs in one connected system.

- Mobile side: Sales Representatives, Territory Distributors, and Shop Owners.
- Web side: Warehouse Managers, Admins, and Demand Planners.
- Core output: structured and traceable data that can be monitored, exported, and used for demand planning.

## 4. User roles and responsibilities

| Role                 | Main platform | Main purpose                                                             | Can create orders?                                  | Key limits                                                       |
|----------------------|---------------|--------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------|
| Sales Representative | Mobile        | Visit shops, collect field data, assign new shops, place orders only for | Yes, but only for new shops or non-app/rural shops. | Does not control warehouse inventory policy.<br>Does not approve |

|                       |        |                                                                                                                  |                                      |                                                                                |
|-----------------------|--------|------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------|
|                       |        | unregistered/rural shops, deliver from carried stock.                                                            |                                      | master data globally.                                                          |
| Territory Distributor | Mobile | Handle assigned deliveries and remaining balances from warehouse-assigned orders.                                | No.                                  | Does not do ordering, OSA checks, or EN reports.                               |
| Shop Owner            | Mobile | Place orders for their own shop and view order history.                                                          | Yes, only for their own linked shop. | No OSA checking. No advanced insight dashboard in final plan.                  |
| Warehouse Manager     | Web    | Issue stock, receive territory orders, record given/returned/damaged items, assign remaining deliveries.         | No direct customer order creation.   | Limited to own territory and operational stock control.                        |
| Admin                 | Web    | Full system control over users, territories, products, warehouses, approvals, reports, and planner-level access. | Indirectly through full access.      | Should not manually approve every small action; uses oversight and exceptions. |
| Demand Planner        | Web    | Use ARS integrations, forecasting tool, and non-POS context insights.                                            | No.                                  | Read/analyze only; no operational field execution.                             |

## 5. Platform and access model

- Sales Representative - mobile app, offline-first where possible.
- Territory Distributor - mobile app, offline-friendly where possible.
- Shop Owner - mobile app, lighter workflow than the rep app.
- Warehouse Manager - web only, limited offline support only for important transaction forms if implemented.
- Admin - web only.
- Demand Planner - web only.

Authentication model in the latest plan: both mobile and web users sign in through an approved account flow. The practical interpretation is that account approval is done by Admin, but OTP

should be sent automatically by the system after approval instead of being manually sent by Admin every time.

## 6. Core business rules everybody must follow

| Rule ID | Final Rule                                                                                                                                                                            |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1      | Territory Distributor is the final role name. Do not use Regional Distributor in diagrams or backlog unless you are quoting an older source.                                          |
| R2      | Shop Owner insight charts were removed. Shop Owners only need ordering, repeat order, order history, payment method, and optional feedback.                                           |
| R3      | Sales Representatives place orders only for newly discovered shops, shops without app access, or rural shops without phones.                                                          |
| R4      | Sales Representatives can assign new shops themselves. This creates a shop record and a Shop ID.                                                                                      |
| R5      | When a Shop Owner later joins the app, they can claim the shop using the assigned Shop ID during signup.                                                                              |
| R6      | OTP is not required when a Sales Representative creates a provisional shop record. OTP is required when a Shop Owner creates or activates their own account.                          |
| R7      | Warehouse Managers control start-of-day issuing of products and product transaction records.                                                                                          |
| R8      | Sales Representatives receive a master daily product allocation instead of entering stock manually.                                                                                   |
| R9      | If a Sales Representative cannot fully deliver because carried stock is insufficient, the order balance moves into the warehouse flow and can be assigned to a Territory Distributor. |
| R10     | Demand Planners only use ARS integrations, non-POS context insights, and forecasting tools.                                                                                           |

## 7. Shop identity and account-linking model

This is one of the most important parts of the final design because it resolves the conflict between rep-created shops and Shop Owner self-service accounts.

### 7.1 Shop record states

- Provisional shop: created by a Sales Representative in the field when the shop is new or not yet digitally connected.
- Assigned shop: a provisional or verified shop already linked to a Sales Representative and territory.

- Claimed shop: the Shop Owner has used the Shop ID to link their account to the existing shop record.
- Verified shop: admin or warehouse-side staff has reviewed and confirmed shop details if required.

## 7.2 Signup question for Shop Owners

Recommended wording: "Has your shop already been registered by a Sales Representative? If yes, please enter your Shop ID."

## 7.3 Why this matters

- It prevents duplicate shop creation.
- It allows reps to start serving new shops immediately.
- It lets Shop Owners join later without breaking continuity in the data.
- It keeps a single shop identity for orders, history, and assignments.

## 8. Order lifecycle (single agreed version)

Every team member should use the same order lifecycle in diagrams, screens, and test cases. This avoids conflicts between rep-led orders, shop-owner orders, warehouse allocation, and distributor fulfillment.

| State                             | Meaning                                                              | Typical owner at this stage                   |
|-----------------------------------|----------------------------------------------------------------------|-----------------------------------------------|
| Created                           | Order is drafted in the system.                                      | Shop Owner or Sales Representative            |
| Submitted                         | Order is confirmed and sent for processing.                          | Shop Owner or Sales Representative            |
| Allocated                         | Warehouse side has accepted and planned the stock handling.          | Warehouse Manager                             |
| In Delivery                       | Someone is actively trying to deliver the order.                     | Sales Representative or Territory Distributor |
| Partially Delivered               | Some quantity was delivered, but some quantity is still outstanding. | Sales Representative                          |
| Pending Balance                   | Undelivered quantity is waiting for warehouse reassignment.          | Warehouse Manager                             |
| Assigned to Territory Distributor | Outstanding balance is assigned for later delivery.                  | Warehouse Manager                             |
| Completed                         | All required quantity has been delivered and confirmed.              | Sales Representative or Territory Distributor |
| Closed                            | Order is finalized for reporting and no more edits are allowed.      | System / Admin rules                          |

### 8.1 Order source rules

- Shop Owner app order: used when the shop already has app access.
- Sales Representative order: used only for new shops, non-app shops, or rural shops without phones.

- Territory Distributor never creates customer orders.

## **8.2 Delivery split rule**

- If carried stock is enough, the Sales Representative can complete delivery directly.
- If carried stock is not enough, the Sales Representative delivers what is available, marks the rest as pending, and the remaining balance enters the warehouse flow.
- Warehouse Manager can assign that remaining balance to a Territory Distributor.

## **9. End-to-end role workflows**

### **Sales Representative workflow**

1. Sign in and access assigned route and shops.
2. Enter or confirm lorry/vehicle identity if needed.
3. View lorry dashboard populated from master daily allocation and warehouse-issued stock.
4. Visit assigned shops and perform store visit flow.
5. Check OSA / shelf availability and related field observations.
6. Create a new shop if the outlet is not yet in the system.
7. Place orders only for new shops, non-app shops, or rural shops without phones.
8. Deliver products from carried stock where possible.
9. If stock is short, perform partial delivery and send the remaining balance into the warehouse process.
10. Close the day with return and reconciliation records.

### **Territory Distributor workflow**

11. Sign in and view assigned deliveries.
12. Collect the exact warehouse-issued quantities for assigned deliveries.
13. Deliver assigned orders or order balances.
14. Record collected cash if applicable.
15. Return remaining products if any remain.
16. Close the distributor day by submitting money and product returns.

### **Shop Owner workflow**

17. Sign up or sign in.
18. If the shop was already registered by a Sales Representative, claim the shop using Shop ID.
19. Place an order or repeat a previous order.
20. Choose payment method such as cash on delivery or card.
21. View order history.
22. Optionally provide feedback after order completion.

### **Warehouse Manager workflow**

23. Start the day for the territory.
24. Set or review master daily allocation rules for Sales Representatives.
25. Issue products to Sales Representatives.

26. Issue exact products to Territory Distributors based on assigned deliveries.
27. Receive submitted orders in the territory queue.
28. Record product transactions: given, returned, damaged, expired, and reconciled quantities.
29. Assign pending balances to Territory Distributors when needed.
30. Track distribution and reconcile daily stock.

### Admin workflow

31. Approve user accounts and manage role assignments.
32. Maintain territories, shops, products, and warehouse-level master data.
33. Monitor all users and all territories.
34. View reports and audit activities.
35. Access planner-level dashboards and full system visibility.

### Demand Planner workflow

36. Sign in to the web dashboard.
37. Use ARS integration exports.
38. Use the forecasting tool for validation and analysis.
39. View non-POS context insights and demand planning dashboards.

## 10. Feature boundaries and non-conflict decisions

This section is important for the use case and activity diagram team. It explains where feature boundaries are now fixed so diagrams do not contradict each other.

| Boundary                                                      | Agreed decision                                                                                                                                                                       |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sales Representative vs Shop Owner ordering</b>            | Sales Representative ordering is only a fallback channel for unregistered, non-app, or rural shops. Once a shop uses the app, normal self-service ordering belongs to the Shop Owner. |
| <b>Sales Representative vs Territory Distributor delivery</b> | Sales Representatives can deliver from their carried stock. Territory Distributors handle exact assigned deliveries and pending balances pushed by the warehouse.                     |
| <b>Warehouse Manager vs Admin master control</b>              | Admin owns global control and master governance. Warehouse Manager owns territory-level operational stock flow.                                                                       |
| <b>Demand Planner vs Admin analytics access</b>               | Demand Planner focuses on forecast-related views. Admin has broader full-access visibility.                                                                                           |
| <b>Shop creation vs shop account creation</b>                 | A shop record can be created by a Sales Representative without OTP. A Shop Owner user account still requires proper account activation.                                               |
| <b>Offline support</b>                                        | Mobile should be offline-first where possible. Web should not promise full offline support except limited offline-friendly transaction handling if implemented.                       |

## 11. Guidance for the use case diagram team

- Use the six final actors only: Sales Representative, Territory Distributor, Shop Owner, Warehouse Manager, Admin, Demand Planner.
- Do not use older actor names such as Regional Distributor in the final diagrams.
- Keep the overall diagram focused on major use cases, not tiny UI actions.
- Treat order lifecycle as shared across the system; do not draw separate contradictory order flows.
- Show Shop Owner linking through Shop ID.
- Show Sales Representative assigning shops.
- Show Warehouse Manager issuing products and assigning pending balances.
- Show Demand Planner only on forecasting/insight/export use cases.

## 12. Guidance for the activity diagram team

- Use swimlanes where necessary because the workflows cross roles frequently.
- The most important activity diagrams are: shop onboarding/linking, shop-owner self-ordering, rep order-and-delivery flow, partial-delivery-to-distributor flow, warehouse start-of-day allocation, and distributor delivery closure.
- Always show decision points for whether the shop already has app access, whether carried stock is enough, and whether an order balance must be assigned to a Territory Distributor.
- Use the agreed order states consistently.

## 13. Guidance for the testing team

Testing should be organized role by role and scenario by scenario rather than screen by screen.

| Test area                       | What must be validated                                                                                                 | Example critical scenarios                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication and linking      | Account approval, OTP flow, Shop ID claiming, role-based access.                                                       | Approved user logs in; Shop Owner claims existing shop using Shop ID; unauthorized role cannot access planner tools.                               |
| Sales Representative field flow | Shop assignment, route access, OSA recording, order creation only for allowed shop types, delivery from carried stock. | Rep creates provisional shop; rep places order for rural shop; rep cannot place standard order for an app-enabled shop if business rules block it. |
| Warehouse flow                  | Issue products, receive returns, reconcile stock, assign pending balances.                                             | Warehouse issues stock to rep; warehouse records damaged returns; warehouse assigns remaining balance to Territory Distributor.                    |
| Territory Distributor flow      | View assigned deliveries, complete delivery, close day, return remaining items.                                        | Distributor receives assigned order balance and completes it.                                                                                      |



|                           |                                                                                  |                                                                                                       |
|---------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Shop Owner flow           | Sign up, claim shop, place order, repeat order, view history, payment selection. | Shop Owner enters Shop ID; repeated order recreates items correctly; history shows past orders.       |
| Planner/reporting flow    | Exports, forecasting tool, non-POS context views.                                | Planner filters by SKU and territory; export file generated correctly.                                |
| Cross-role data integrity | State transitions and single-source truth.                                       | Partially delivered order becomes pending balance, then assigned, then completed without duplication. |

## 14. Suggested MVP vs extension split

This split makes the project more buildable and helps the team prioritize.

| Priority  | Include                                                         | Notes                                                                         |
|-----------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| MVP       | Six-role login/access model                                     | Role visibility can still be simplified in UI, but the logic should be final. |
| MVP       | Rep shop assignment and Shop ID linking                         | This resolves a major system conflict.                                        |
| MVP       | Shop Owner ordering + order history                             | Insight charts removed from final plan.                                       |
| MVP       | Warehouse start-of-day issue and return records                 | Core operational traceability.                                                |
| MVP       | Rep delivery + partial delivery + pending balance flow          | Critical business flow.                                                       |
| MVP       | Territory Distributor assigned delivery flow                    | Needed to resolve shortages.                                                  |
| MVP       | Admin monitoring and master data                                | Needed to run the system.                                                     |
| MVP       | Demand Planner exports + forecast validation + non-POS insights | Core planning purpose.                                                        |
| Extension | Limited offline web support                                     | Nice to have if time allows.                                                  |
| Extension | Advanced approvals and exception queues                         | Can be added if implementation time remains.                                  |
| Extension | QR or barcode enhancements                                      | Good for polish, not essential to final logic.                                |
| Extension | Extra audit analytics and dashboards                            | Useful but not required for basic system coherence.                           |

## 15. Product backlog (all major features)

This backlog is grouped by role and system area so the whole team can trace features easily. It is intentionally broad; your sprint planning can then prioritize from it.

### **Shared / platform-wide**

- User registration and sign-in
- OTP verification after approval
- Role-based access control
- Territory-based visibility control
- Audit logging
- Notification handling
- Offline sync for mobile users
- Status-driven order lifecycle
- Search and filtering

### **Sales Representative app**

- View assigned shops
- View daily route
- Store visit/check-in flow
- OSA and shelf availability capture
- Expiry / damage flag capture
- Assign new shop
- Generate or store Shop ID
- Place order for unregistered/rural/non-app shop
- Deliver from carried stock
- Partial delivery handling
- End-of-day return and reconciliation
- Lorry dashboard / carried stock view

### **Shop Owner app**

- Sign up
- Sign in
- Claim assigned shop using Shop ID
- Place order
- Repeat previous order
- View order history
- Choose payment method
- Provide optional feedback

### **Territory Distributor app**

- View assigned deliveries
- Deliver assigned orders
- Record collected cash
- Return remaining products
- Close distributor day

### **Warehouse Manager web**

- Start the day
- Set master daily product allocation
- Issue products to Sales Representatives
- Issue products to Territory Distributors
- Receive territory orders
- View territory order queue
- Record products given
- Record products returned
- Record damaged products
- Record expired products
- Track product distribution data
- Assign remaining delivery to Territory Distributor
- Reconcile daily stock

### **Admin web**

- Approve user accounts
- Manage user roles
- Manage territories
- Manage shop master data
- Manage products/SKUs
- Manage warehouses
- Monitor all users
- Monitor all territories
- View reports
- View audits and exception records
- Full dashboard access

### **Demand Planner web**

- Access ARS integration exports
- Generate export files
- View export history
- Use forecasting tool
- Filter by SKU/territory/date
- View forecast vs actual
- View non-POS context insights
- View demand planning dashboard

### **Data and reporting**

- Sales feed export
- Inventory feed export
- Orders feed export
- Master data feed export

- Service-level reporting
- Territory performance reporting
- Backorder / pending balance reporting
- Returns and stock mismatch reporting

## **16. Final reminder to the whole group**

Whenever someone creates a diagram, screen, test case, backlog ticket, or database table, they should compare it against this document first. The main thing to protect is consistency: who owns the shop, who creates the order, who fulfills it, who controls stock, and who uses the final data.

*Prepared as the shared final-plan reference for Group 06*